Roll No. ....

TPE–016

B. TECH. (ECE)
(EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

CMOS ANALOG CIRCUIT DESIGN

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Derive expression for current voltage in MOS transistor. (CO1)

OR

(b) Explain in detail about small signal model of MOS transistor. (CO1)

2. (a) What do you understand by MOS device capacitance ? (CO1)

OR

(b) Explain role of threshold voltage in Depletion Mode Enhancement mode MOS transistor. (CO1)

3. (a) What do you understand about common source resistive load single stage amplifier ? (CO2)

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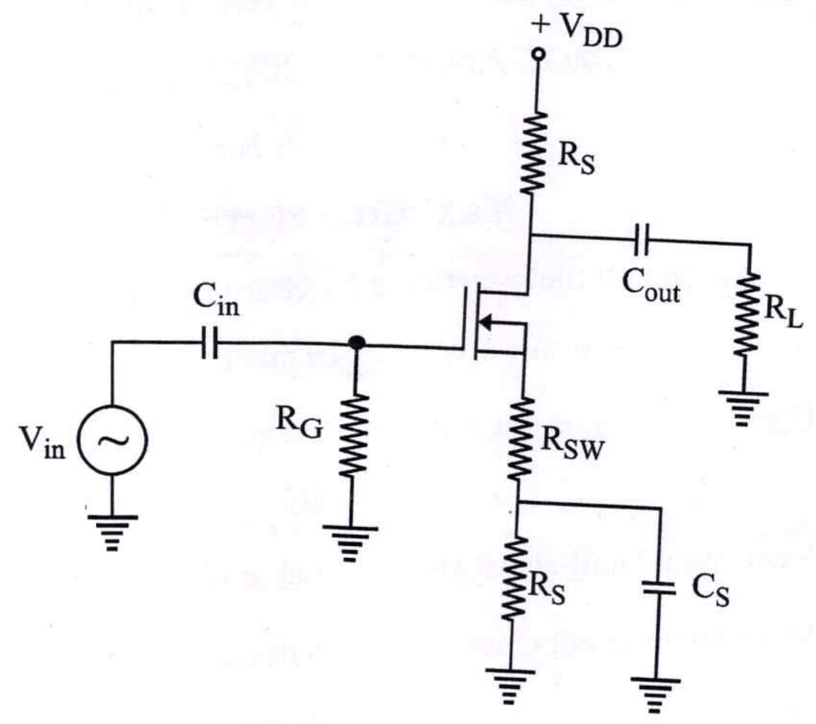
OR

(b) What do you understand about CS with diode connected load single stage amplifier ? (CO2)

4. (a) Derive expression for voltage gain MOSFET Common Source Amplifier. (CO2)

OR

(b) For the amplifier in Figure, determine the input impedance and load voltage. V_{in} = 20 mV, V_{DD} = 20 V, R_G = 1 M Ω, R_D = 1.8 kΩ, R_{SW} = 20 Ω, R_S = 400 Ω, R_L = 12 k Ω, I_{DSS} = 40 mA, $V_{GS(off)}$ = – 1V. (CO2)



5. (a) Explain the working of MOS transistors based differential amplifier. (CO2/CO3)

OR

(b) Explain source follower CG single stage amplifier. (CO2/CO3)

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TPE–008

B. E. (ECE) (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

NEURAL NETWORK AND MACHINE LEARNING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain the structure of a biological neuron and illustrate the model used for neural computing. (CO1)

OR

(b) Explain McCulloch Pitts model of neural network and illustrate implementation of AND operation. (CO1)

2. (a) Design a pattern classifier using basic neural network model with S as a set of patterns to be classified into classes A and B : (CO1)

S = {000, 010, 010, 011, 100, 101, 110, 111}

A = {000, 001, 010, 100}.

B = {011, 101, 110, 111}

OR

(b) Illustrate implementation using neural network of the following table :

(CO1)

(a) Truth Table		
x_1	x_2	x_1 **AND (NOT** x_2**)**
0	0	0
0	1	0
1	0	1
1	1	0

3. (a) Explain the structure of Perceptron. Implement OR Operation using the Perceptron : (CO1)

OR

(b)

X1	**X2**	**Y**
0	0	0
0	1	0
1	0	0
1	1	1

Implement a Neural Network to implement the truth table. (CO1)

4. (a) What are the different types of learning schemes for neural network ? Explain with examples. (CO2)

OR

(b) With Hebb Learning Rule train the neural network to realise the following table : (CO2)

Training set			
Input Patterns			Output
x_0	x_1	x_2	t
+ 1	1	1	1
+ 1	1	− 1	− 1
+ 1	− 1	1	− 1
+ 1	− 1	− 1	− 1

5. (a) What are Supervised and Unsupervised Machine Learning techniques ? Different them with the help of a suitable example. (CO2)

OR

(b) Explain any *two* activation functions. (CO2)

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TEC–801

B. TECH. (EC) (EIGHT SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

COMPUTER ARCHITECTURE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Tabulate the difference between the following : (CO1)

    (i) RISC and CISC

    (ii) Von-Neumann and Harvard Architecture

OR

   (b) Explain the various components of a computer system using suitable diagrams. (CO1)

2. (a) With the help of suitable example explain the concept of memory addressability. (CO1)

OR

   (b) Give the 32-bit and 64-bit IEEE floating-point representation of the following numbers : (CO1)

    (i) – 23.125

    (ii) + 67.5

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3. (a) What is an addressing mode ? Explain the various addressing modes with suitable examples. (CO2)

OR

(b) Draw the flowchart and explain the Booth's algorithm using suitable examples. (CO2)

4. (a) Describe non-restoring division with an example. (CO2)

OR

(b) Add the numbers $(0.75)_{10}$ and $(-0.275)_{10}$ in binary using the floating-point addition algorithm. (CO2)

5. (a) Explain how the computer executes a program. (CO1)

OR

(b) Define the following terms : (CO1)

(i) Compiler

(ii) Mnemonics

(iii) Opcode

(iv) Machine Cycle

(v) Datapath

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TPE–014

B. TECH. (ECE) (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

TESTING FOR VLSI CIRCUIT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the importance of CAD tools in VLSI. (CO1)

OR

(b) Explain VLSI technology trends affecting testing in detail. (CO1)

2. (a) Explain the role of testing and types of testing. (CO2)

OR

(b) Define the term controllability and observability. Describe the gate propagation rules for combinational controllability. (CO2)

3. (a) Explain the following test with respect to electrical testing : (CO1)

(i) Power Consumption Test

(ii) Output Short Current Test

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OR

(b) Explain the term "Fault equivalence" and "fault collapsing." Mention the importance of fault collapsing. (CO1)

4. (a) Explain stuck at faults in detail with proper examples. (CO1)

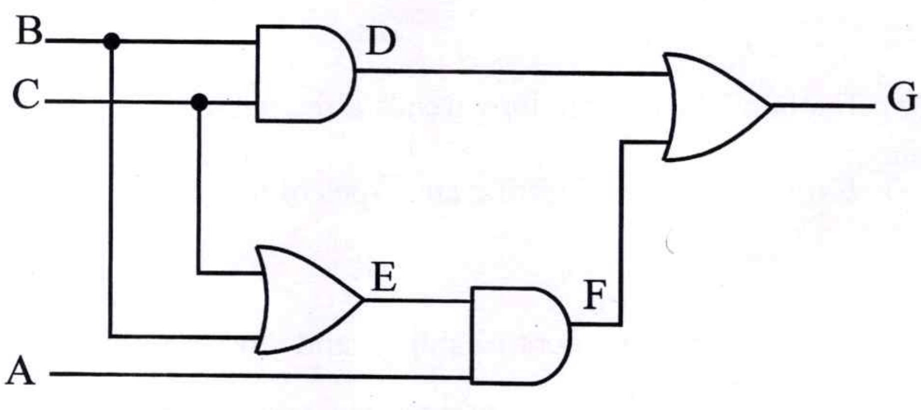
OR

(b) Explain the term "simulation for design verification and test evaluation." (CO1)

5. (a) Explain the difference between defects, errors and faults with an example. (CO2)

(

(b) For a digital circuit given be ... method to generate the test vector for node E to be stuck at 0 ? (CO2)



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TPE–015

B. TECH. (ECE) (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

WIRELESS SENSOR NETWORKS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain the basic operating principle of wireless adhoc networks. (CO2)

OR

(b) Point at least *ten* differences between cellular networks and adhoc wireless networks. (CO2)

2. (a) How is Wireless Sensor Networks (WSN) is subset of wireless mobile adhoc networks ? Explain their characteristics. (CO1)

OR

(b) Explain the unique constraint and challenges of Wireless Sensor Networks (WSN). (CO1)

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3. (a) Explain at least *five* applications of WSN in detail. (CO1/CO2)

OR

(b) What are the specific issues that makes the sensor networks a distinct category of adhoc wireless networks ? (CO1/CO2)

4. (a) How is adhoc networks useful in military and emergency applications ? (CO2/CO3)

OR

(b) Explain the "Hidden Terminal Problem" and "Exposed Terminal Problem" in adhoc wireless networks. (CO2/CO3)

5. (a) Explain the "Hybrid Network" in MANET. (CO1/CO3)

OR

(b) How do MACA (Multiple Access Collision Avoidance) protocol reduce the Hidden and Exposed Terminal Problems ? (CO1/CO3)